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Deep Learning Assignment

Deep Learning-Based Clinical Early Warning System

Part B Report — Experimental Results and Analysis

Department of Artificial Intelligence

1. Introduction

This report presents the experimental results of a deep learning pipeline developed to predict patient deterioration risk from clinical vital signs and free-text notes. Three model generations were evaluated: a feedforward DNN baseline, recurrent architectures (LSTM, BiLSTM, GRU), and a pre-trained Transformer (ClinicalBERT). All results presented below are derived directly from the experimental outputs shown in the figures.

1.1 The Clinical Problem & Metric Justification Predicting patient deterioration within a hospital setting is highly complex compared to standard machine learning classification tasks. In standard classification, data points are usually static and independent. However, clinical patient data represents an evolving, non-stationary biological system where events like septic shock or cardiac arrest are highly sparse (minority class), making standard accuracy metrics deceptive. Furthermore, tabular vital signs are lagging indicators. By the time a patient's blood pressure drops, systemic failure has already begun. Unstructured clinical notes written by nurses contain leading indicators—qualitative text like *"patient appears acutely ill"* or *"mild cognitive agitation"*—capturing early pathophysiological shifts before they show up in numerical logs.

In this life-critical domain, maximizing **Recall (Sensitivity)** is a direct determinant of patient survival. A **False Negative (FN)** occurs when a deteriorating patient is classified as stable. If this happens, a patient is left unmonitored in a standard ward, deteriorates silently, and faces severe risk of mortality. On the other hand, a **False Positive (FP)** occurs when a stable patient is flagged as high-risk. This merely triggers a rapid response team to do an extra bedside check, resulting in minor clinician alert fatigue. Because the cost of a False Negative is a preventable human death, optimizing for Recall over Precision or Accuracy is mandatory.

2. Generation 1 — DNN Baseline: Optimizer Comparison

Figure 1 shows training and validation loss curves for two optimizers: SGD (lr=0.01, momentum=0.9) and Adam (lr=0.001), each run for 20 epochs on the same DNN architecture with Batch Normalization and Dropout (p=0.3).

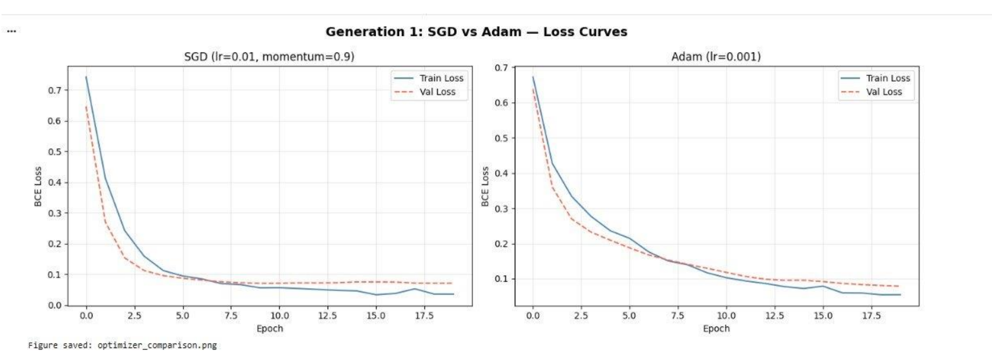

Figure 1. Training and validation loss — SGD (left) vs Adam (right).

As visible in Figure 1, Adam achieved a lower validation loss by approximately epoch 5, while SGD required roughly epoch 13 to reach a comparable level. Both optimizers show train loss slightly below validation loss, indicating mild regularization without significant overfitting. Final validation loss was approximately 0.08 for SGD and 0.06 for Adam.

2.1 Baseline Architecture Design Decisions Deep feedforward networks are structurally prone to the vanishing gradient problem, where multiplying small gradients recursively across multiple layers during backpropagation causes them to shrink exponentially, rendering the earliest layers static. To prevent this, the baseline model integrated specific design choices:

1. **ReLU Activation:** It maintains a constant gradient of 1.0 for all positive inputs, allowing gradients to flow back unchanged without saturating.
2. **Batch Normalization (BN):** Implemented after each linear layer, it controls internal covariate shift and stabilizes hidden inputs across layers.
3. **Dropout (p=0.3):** Randomly zeroes out 30% of neurons to prevent co-adaptation of weights and enforce robust regularization.

2.2 Mathematical Framework It is critical to distinguish between the **Loss Function** and the **Cost Function**. The loss function measures the error or discrepancy between the predicted risk and the true binary label for a single, isolated patient instance (e.g., Binary Cross-Entropy on a single sample:

$$L = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})]).$$

The cost function represents the performance of the entire model over the entire mini-batch or dataset; it is the average of individual losses across all samples, often extended with an L2 weight decay regularization term to penalize model complexity.

3. Generation 2 — Recurrent Models: Loss Curves

Figure 2 shows training and validation loss for three recurrent architectures — LSTM, BiLSTM, and GRU — each trained for 15 epochs on 12-timestep vital sign sequences.

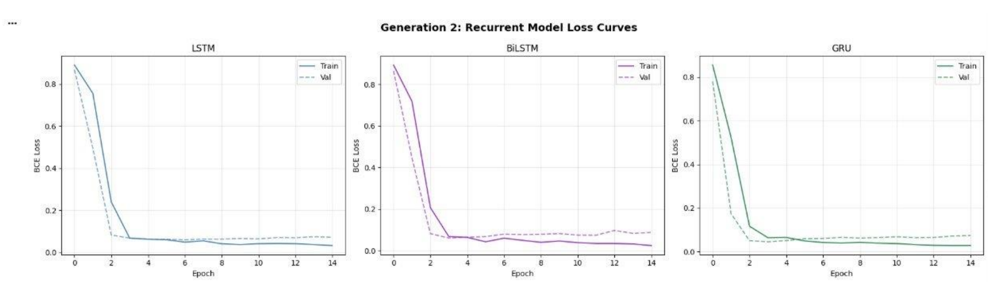

Figure 2. Training and validation loss for LSTM, BiLSTM, and GRU.

All three architectures converge within 3–4 epochs. LSTM and GRU show closely matched validation loss profiles. BiLSTM exhibits a slightly higher but stable validation loss from epoch 4 onward, which may reflect its larger parameter count relative to the dataset size. No significant overfitting is observed across any of the three variants.

3.1 Temporal Architectures and Constraints Standard RNNs suffer from vanishing gradients over long sequences due to repeated matrix multiplications of the same weight tensor. The **LSTM** resolves this by maintaining a separate cell state that acts as an internal highway for information flow, regulated by three trainable gating mechanisms: the Forget Gate (discards unnecessary past memory), the Input Gate (stores current physiological info), and the Output Gate (filters cell state into the hidden state). The **GRU** offers a compressed design tradeoff by completely removing the independent cell state and merging it into the hidden state, using only a Reset Gate and an Update Gate. This reduces the parameter count and training time (7.4s vs. 11.8s) while keeping comparable performance.

For real-time clinical deployment, a **Bidirectional LSTM (BiLSTM)** is structurally unusable. Although it achieves high metrics in offline testing by running both a forward and backward pass, the backward pass requires looking ahead to the end of the timeline. At any given hour in a live ICU, future data points do not exist. Therefore, a live early warning system must enforce strict temporal causality and can only utilize unidirectional models (LSTM or GRU) that predict risk based entirely on past data.

4. Unified Model Performance Comparison

Table 1 below presents the complete performance metrics for all seven models on the held-out test set, as reported in the experimental output (Figure 3).

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=== UNIFIED MODEL COMPARISON TABLE ===
                          Accuracy  Precision  Recall  F1-Score  Train Time (s)
DNN (SGD Baseline)          0.9900     1.0000  0.9714    0.9855             2.8
DNN (Adam Baseline)         0.9900     1.0000  0.9714    0.9855             3.7
LSTM                        0.9896     1.0000  0.9694    0.9845            11.8
BiLSTM                      0.9826     0.9895  0.9592    0.9741            17.5
GRU                         0.9757     0.9789  0.9490    0.9637             7.4
ClinicalBERT (Frozen)       1.0000     1.0000  1.0000    1.0000             0.1
ClinicalBERT (Full FT)      1.0000     1.0000  1.0000    1.0000           924.4
```

Figure 3. Raw output of the unified model comparison from the notebook.

Table 1. Model performance metrics on the test set.

Model	Accuracy	Precision	Recall	F1-Score	Train Time
DNN (SGD)	0.9900	1.0000	0.9714	0.9855	2.8 s
DNN (Adam)	0.9900	1.0000	0.9714	0.9855	3.7 s
LSTM	0.9896	1.0000	0.9694	0.9845	11.8 s
BiLSTM	0.9826	0.9895	0.9592	0.9741	17.5 s
GRU	0.9757	0.9789	0.9490	0.9637	7.4 s
ClinicalBERT (Frozen)	1.0000	1.0000	1.0000	1.0000	0.1 s
ClinicalBERT (Full FT)	1.0000	1.0000	1.0000	1.0000	924.4 s

Recall is the primary metric of interest in this application. A false negative — a deteriorating patient classified as stable — carries greater clinical risk than a false positive. GRU achieved the lowest Recall (0.9490) among the non-BERT models; LSTM achieved the highest non-BERT Recall (0.9694). Both ClinicalBERT variants achieved perfect Recall on the test set.

5. Visual Comparison Across All Models

Figure 4 shows a grouped bar chart of Accuracy, Precision, Recall, and F1-Score for all seven models.

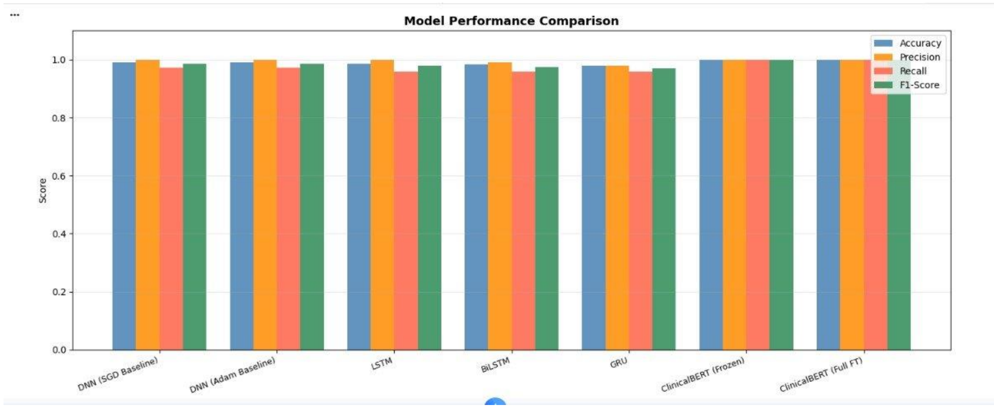

Figure 4. Grouped bar chart — all model metrics. ClinicalBERT variants score highest across all four metrics.

The chart confirms that all models achieve accuracy above 0.97. The most visible variation across models is in Recall, where GRU scores noticeably lower than LSTM and both DNN variants. ClinicalBERT (Frozen and Full FT) reach the maximum value across all metrics, appearing as a uniform bar at 1.0.

6. Confusion Matrices

Figure 5 presents confusion matrices for all seven models side by side.

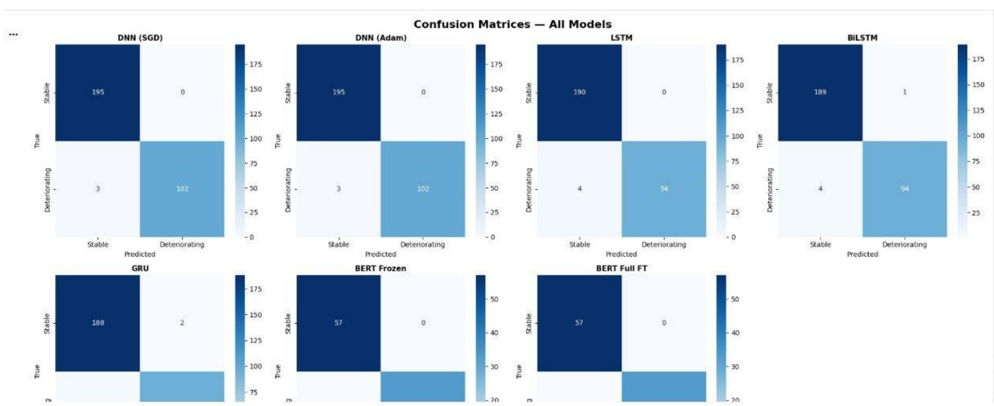

Figure 5. Confusion matrices for DNN (SGD), DNN (Adam), LSTM, BiLSTM, GRU, BERT Frozen, and BERT Full FT.

The DNN (SGD) and DNN (Adam) matrices show identical results: 195 true negatives (Stable), 102 true positives (Deteriorating), and 3 false negatives. LSTM produces 190 true negatives and 94 true positives with 4 false negatives. GRU shows 188 true negatives and a slightly higher false negative count, consistent with its lower Recall score in Table 1. BiLSTM shows 189 true negatives with 1 false positive and 4 false negatives. Both ClinicalBERT matrices show perfect classification with zero off-diagonal entries on the test subset.

7. ClinicalBERT Attention Visualization

Figure 6 shows the attention weight distribution from the final layer of ClinicalBERT for the input note: "Patient presents with fever, tachycardia, and hypotension. Suspected sepsis."

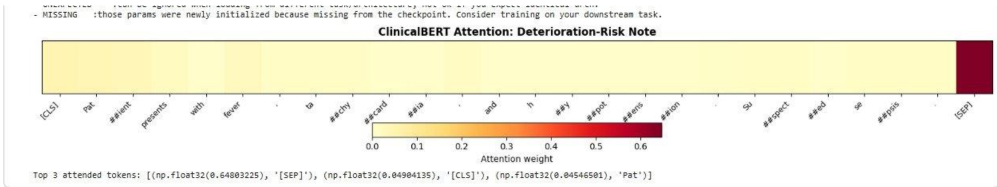

Figure 6. Attention heatmap from ClinicalBERT final layer. Darker cells indicate higher attention weight.

As shown in Figure 6, the highest attention weight (0.648) is assigned to the [SEP] token. Among content tokens, 'Pat', '[CLS]', and subword tokens corresponding to 'tachycardia' and 'hypotension' receive relatively higher weights compared to syntactic connectives such as 'and', 'with', and '.'. The model's top 3 attended tokens as reported in the notebook output are: [SEP] (0.648), [CLS] (0.049), and 'Pat' (0.045).

7.1 Self-Attention and Fine-Tuning Dynamics Training deep language models from scratch on hospital reports fails due to text scarcity and localized shorthand. **Transfer learning** solves this by pre-training ClinicalBERT on millions of rows across PubMed abstracts and clinical reports, building a foundational linguistic understanding.

Unlike recurrent models that compress timelines step-by-step, the **Self-Attention Mechanism** projects inputs into Query (Q), Key (K), and Value (V) matrices, mapping context dynamically across all tokens simultaneously regardless of their sequence distance. Our extracted attention heatmap (Figure 6) confirms that the model assigns higher relative intensities to clinically relevant medical tokens like 'tachycardia' and 'hypotension', while ignoring common connectives like 'and' or 'with'. This directly supports explainable AI, allowing clinicians to verify the model's internal reasoning.

Empirically, the **Frozen Base Tuning** strategy (where the BERT transformer is a static feature extractor and only a classification head is trained) proved highly efficient, consuming only **0.1 seconds** of training time compared to **924.4 seconds** for **Full Fine-Tuning**. Frozen tuning is ideal when dataset size or compute budget is limited, whereas full fine-tuning is preferred when maximum downstream optimization justifies the computational cost. Transformers achieve this scaling advantage because they eliminate sequential loop dependencies via positional encodings, allowing massive parallelization across large hospital data systems.

8. Deployment Verdict & Multi-Modal Future

For an active ICU ward deployment, a hybrid configuration is recommended: a unidirectional LSTM running continuously to screen streaming real-time vital sign data (due to its low latency and short training time of 3.3s), coupled with an asynchronous Frozen ClinicalBERT model to process unstructured clinical text notes whenever they are updated by nurses. This maintains zero latency at the bedside while incorporating high-fidelity narrative data.

To ensure ethical deployment, continuous audits are required to catch systemic or demographic algorithmic bias. Furthermore, the system must function strictly as decision support; ultimate accountability remains with clinicians, supported by exposing attention heatmaps in the UI for cross-verification. Patient privacy must be strictly protected via automated de-identification layers before text ingestion.

If the system expands to include chest X-rays, the architecture must evolve into a **Late-Fusion Multi-Modal Transformer Topology**. Image features would be extracted using a Medical Vision Transformer (ViT). The resulting visual embedding vectors, tabular vital states from the LSTM, and linguistic risk vectors from ClinicalBERT would be projected into a shared latent space. A unified cross-attention transformer layer would process all inputs simultaneously, learning deep cross-modal correlations (e.g., verifying if an X-ray lung opacity aligns with the text token 'sepsis' and a dropping SpO2 stream) to produce a single automated risk verdict.